

3.1 Introduction

Now that you understand the fundamentals, tools, and preparation, this module focuses on the actual execution of grafting techniques. These are the most widely used and practical methods in both small-scale gardening and commercial production.

This module will guide you through:

Whip (Splice) Grafting

Cleft (Wedge) Grafting

Bud Grafting (Budding)

Each technique is used under different conditions depending on plant size, age, and purpose.

3.2 Understanding Cambium Alignment (Core Principle)

Before performing any technique, remember:

The success of any graft depends on proper cambium alignment.

Cambium = thin, living layer responsible for growth

Must be in direct contact between scion and rootstock

Even partial alignment can lead to success

Without this, the graft will fail—regardless of technique.

3.3 Whip (Splice) Grafting

Best Used When:

Scion and rootstock are similar in diameter

Typically young plants (seedlings or small branches)

Step-by-Step Procedure

Step 1: Prepare the Rootstock

Make a slanted cut (about 1–2 inches long)

Ensure the cut is smooth and even

Step 2: Prepare the Scion

Make an identical slanted cut at the base

Match the angle and length of the rootstock cut

Step 3: Join the Pieces

Align the cambium layers carefully

Fit the cuts together like puzzle pieces

Step 4: Secure the Union

Wrap tightly with grafting tape

Ensure no movement

Step 5: Seal the Graft

Apply wax or sealant to prevent drying

Optional Advanced Variation: Whip-and-Tongue

A small “tongue” cut is added to both parts

Provides better stability and stronger union

Advantages

Strong and secure union

High success rate when done properly

Fast healing

3.4 Cleft (Wedge) Grafting

Best Used When:

Rootstock is thicker than the scion

Used for topworking or changing plant varieties

Step-by-Step Procedure

Step 1: Prepare the Rootstock

Cut the branch straight across

Split the center vertically (creating a cleft)

Step 2: Prepare the Scion

Shape the base into a wedge (V-shape)

Step 3: Insert the Scion

Place the wedge into the cleft

Align cambium on at least one side

Step 4: Secure the Union

Wrap firmly to hold the scion in place

Step 5: Seal All Exposed Areas

Apply grafting wax to prevent drying and infection

Advantages

Ideal for older or mature trees

Allows changing varieties without replanting

Can insert multiple scions in one rootstock

Important Note

Because the sizes differ, perfect alignment on both sides is not required—focus on aligning at least one side.

3.5 Bud Grafting (Budding)

Best Used When:

Limited scion material is available

Plants with active bark (easy to peel)

Common in fruit trees and roses

Concept

Instead of a full stem, a single bud is used as the scion.

Step-by-Step Procedure (T-Budding Example)

Step 1: Prepare the Rootstock

Make a T-shaped cut in the bark

Gently lift the bark flaps

Step 2: Prepare the Bud

Slice a bud from the scion with a thin layer of wood

Step 3: Insert the Bud

Slide the bud into the T-cut under the bark

Step 4: Wrap the Area

Secure with grafting tape

Leave the bud exposed

Aftercare

Once the bud starts growing, cut back the rootstock above it

This directs energy into the new shoot

Advantages

Efficient use of plant material

High success rate

Common in commercial nurseries

3.6 Comparison of Techniques

Technique	Best For	Skill Level	Key Benefit
Whip Graft	Same-size plants	Beginner	Strong, fast-healing union
Cleft Graft	Thick rootstock	Intermediate	Useful for mature trees
Budding	Limited material, active growth	Intermediate	Efficient and widely used

3.7 Common Mistakes in Execution

Avoid these errors during grafting:

Misalignment of cambium layers

Loose wrapping causing movement

Cuts that are rough or uneven

Letting scion dry before joining

Delayed wrapping after cutting

3.8 Signs of a Successful Graft

Scion remains green and healthy

Buds begin to grow

Callus forms at the union

No signs of drying or decay

3.9 Signs of Graft Failure

Scion becomes dry or black

No new growth after several weeks

Weak or broken union

Rootstock produces unwanted shoots

3.10 Practical Demonstration Scenario

Example: Grafting a Fruit Tree

Select a healthy rootstock (disease-resistant)

Choose a scion with high-quality fruit traits

Use whip graft for young plants or cleft graft for older trees

Secure and seal the graft

Monitor growth and remove unwanted shoots

3.11 Learning Outcomes

By the end of this module, you should be able to:

Perform whip, cleft, and bud grafting techniques

Choose the appropriate grafting method for different situations

Align cambium layers correctly

Secure and seal graft unions properly

Identify early signs of success or failure